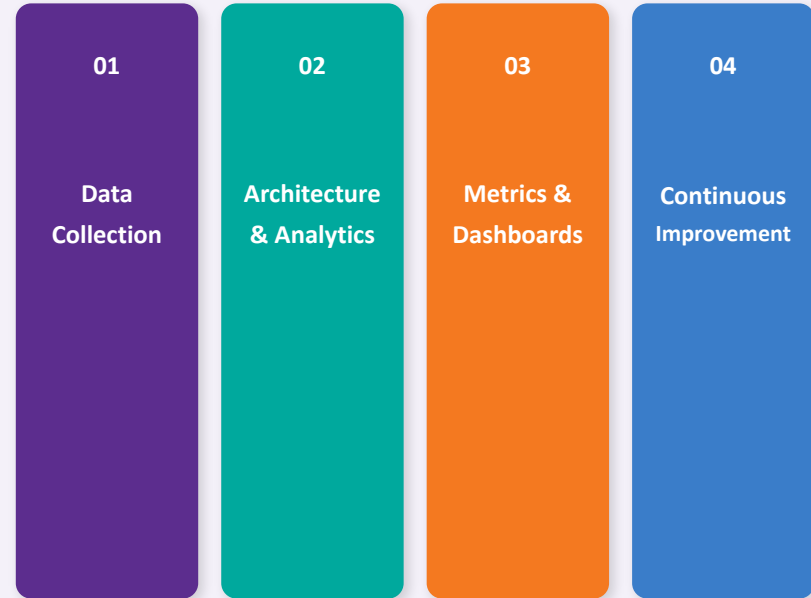


Learning Analytics Framework

Reusable Platform for Measuring,
Monitoring, and Improving Student Outcomes

*First Use Case: Computational Thinking
& Digital Competencies*

Fahad Ali · July 30, 2026



01 · Data Collection, Quality & Measurement Strategy



Core principle: measure BOTH growth from baseline and attainment of the expected competency level not only performance.



Entry Diagnostic / SIS

Baseline for every student before program starts. Measures digital literacy & computational thinking .



Canvas LMS Activity

Automatic daily logs: logins, time on task, content accessed, submissions, discussion activity.



Course Assessments

Quiz scores, assignment grades, scores connected via LTI from any assessment tool.



Structured Project Evaluations

Standardized 4-dimension rubric across all programs. Same criteria for CS and Humanities students.



Student Surveys

5-question pulse survey at entry, midpoint & end. Captures confidence and perceived growth.

DATA QUALITY RULES

- ✓ Data is complete, consistent, Non redundant,
- ✓ No missing data
- ✓ GDPR by design in analytics layer

02 · Data Architecture & Analytics Approach

Reusable Learning Analytics & Measurement Framework configure once for computational thinking, reuse for any future IT:U program

COLLECT

Canvas · Simovative
Assessments · Surveys

RAW ZONE

Untouched source data
Audit trail & GDPR backup

CLEAN ZONE

IDs matched across systems
Dates normalized · Gaps flagged

ANALYTICS ZONE

Universal Student Record
Growth scores · Risk flags

↓ ANALYTICS ENGINE

DESCRIPTIVE

What is happening now?

How many students are at each level?
What is average growth so far?

DIAGNOSTIC

Why is it happening?

Which background groups struggle most?
Which modules have lowest completion?

PREDICTIVE

What will happen next?

Which students are at risk of falling behind?
Based on week 1-4 activity patterns

03 · Key Metrics & Dashboards

Progress Metrics

- Growth Score: current level minus entry baseline
- Assessment improvement rate since entry diagnostic
- Project quality trajectory across the program
- Competency level distribution across cohort

Engagement Metrics

- Weekly active days in Canvas (< 2 days = warning)
- Module completion rate per student per course
- Assignment submission timing (on time vs late)
- Time on task per module too low = skipping content

Program Effectiveness

- Cohort proficiency rate by end of program
- Background group comparison (CS vs Humanities etc)
- Intervention effectiveness did support help?
- Retention rate across full program

THREE DASHBOARD VIEWS

Leadership Dashboard

Weekly

Program health score · Cohort progress trend ·
Background group comparison · At-risk student count

Faculty & Advisors Dashboard

Daily

Course completion heatmap · At-risk student list ·
Assessment distribution · Engagement trend · Module
difficulty flags

Students Dashboard

Real-time

My growth journey (before vs now) · My engagement this
week · My next recommended steps

04 · Using Insights to Continuously Improve Student Outcomes

 Data should change what happens NEXT not just describe what happened BEFORE

01 Early Warning System

Triggered when 2 of 3 conditions are true:

- combine multiple signals
- flag risk is not a diagnosis it trigger human conversation
- human conversation within an agreed timeframe

- Advisor notified within 48 hours for human conversation
- Outcome tracked over following 2 weeks

02 INTERVENTION EVALUATION

Based on current development level:

- Track what happened after support:
- engagement / performance change
 - intervention uptake
 - false positives / missed cases

03 Semester Review Cycle

End of every semester Program Effectiveness Report:

- Which groups progressed most and least?
- Which modules had lowest engagement?
- Which teaching approaches correlated with strongest growth?

- Faculty agree on 2-3 concrete changes for next semester
- Changes measured in following cohort

04 Program Design Loop

Across multiple cohorts evidence base builds:

- If Humanities students consistently struggle week 3-4 of module X
- That is a module design issue, not a student issue
- Data drives curriculum redesign decisions
- IT:U programs improve with every cohort
- General framework applies to any new program

Summary · Why a Reusable Framework

What This Framework Delivers

01

Measures GROWTH from personal baseline fair across CS, Humanities, Arts, Business, and Social Sciences

02

Monitors continuously weekly flags, monthly checkpoints, midpoint diagnostic never waiting until the end

03

Three targeted dashboards for leadership, faculty, and students each sees what they need to act

04

Closes the loop data drives real interventions, semester reviews, and curriculum improvement every cohort

The Reusable Framework Advantage

Configure once, reuse for any future IT:U program

Computational thinking is Use Case 1 not a one-time build

Adding a new program takes days not months

Built on Canvas Data 2 + Simovative SIS systems you already have

GDPR-compliant by design from day one

Gets smarter with every cohort of data